

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (E&C) (Credit System) Degree Examinations**

May - June 2023

20ECE32 – AUTOMOTIVE ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Explain the ignition circuit of an inductive ignition system with the help of a neat block diagram.	5	L*2	1	1
	b) Explain how a throttle angle sensor works with the help of a neat sketch.	5	L2	2	1
	c) Explain the measuring principle of air flow rate sensor with a neat calibration curve and circuit configuration.	5	L2	2	1
	d) With a neat block diagram of Electronic Stability Program control loop, explain the ESP system in a vehicle.	5	L2	1	1
2.	a) Explain the function modules of an electronic system with the help of a neat block diagram.	6	L2	1	1
	b) Explain dual, quad and six headlamp systems.	4	L2	1	1
	c) Illustrate the working principle of knock sensor.	5	L2	2	1
	d) With a neat sketch, explain how to detect shaft position using optical technique.	5	L2	2	1
3.	a) Explain the operation of Adaptive Cruise Control with a neat diagram.	5	L2	1	1
	b) Illustrate the operation of magnetic reluctance position sensor.	5	L2	2	1
	c) Explain the operation of anti-lock braking system with a neat diagram.	5	L2	1	1
	d) With a neat switching characteristic, explain the operation of exhaust gas oxygen sensor.	5	L2	2	1

Unit – II

4.	a) List the different bus systems with one suitable application for each. Explain the technical selection criteria which are considered while selecting a bus system.	4	L2	3	1
	b) Explain the working of full CAN module with a neat diagram.	6	L2	3	1
	c) Explain the AUTOSAR layered software architecture.	10	L2	4	1
5.	a) Explain the message frame format of LIN bus system with a neat sketch of the format.	5	L2	3	1
	b) Explain with a neat diagram, the protocol layers of CAN bus system.	5	L2	3	1
	c) With a neat block diagram, explain the function of AUTOSAR system.	10	L2	4	1
6.	a) Explain the message frame format of CAN bus system with a neat sketch of the format.	5	L2	3	1
	b) List the important characteristics of LIN bus system.	5	L1	3	1
	c) Explain the different interfaces used in defining the ports of software components and BSW modules in AUTOSAR.	6	L2	4	1
	d) Explain the important layers of AUTOSAR basic software.	4	L2	4	1

P.T.O.

Unit – III

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|----|----|---|----|----|---|---|
| 7. | a) | With a basic functional structure, explain the different levels in the vehicle motion domain. | 10 | L2 | 5 | 1 |
| | b) | Explain the functional architecture of motor vehicle system with its open and closed loop subsystems. | 10 | L2 | 5 | 1 |
| 8. | a) | Explain the following terms with respect to data processing in automotive systems. | | | | |
| | | i) Input signals | | | | |
| | | ii) Signal conditioning | | | | |
| | | iii) Signal processing | | | | |
| | | iv) Output signals | 10 | L2 | 5 | 1 |
| | b) | With a neat diagram, explain the architecture of a microcontroller. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
